

Philip Bulsink

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SUMMARY

Senior analytical chemist with 10+ years of experience integrating advanced chromatography, spectroscopy, chemometrics, and scientific software development to solve complex characterization challenges in bioenergy and petroleum research. Recognized for leading international interlaboratory studies, developing novel analytical methods, and translating scientific findings into strategic advice for government and industry stakeholders.

SCIENTIFIC LEADERSHIP & IMPACT

- Principal investigator and technical lead for analytical method development supporting federal bioenergy and fuels research programs.
- Invited technical expert within IEA Bioenergy Task 34, supporting international analytical method development and harmonization for biomass liquefaction oils.
- Lead organizer of international interlaboratory studies involving multiple countries and laboratories.
- Developer of open-source scientific software for chromatographic and spectroscopic data analysis and visualization.
- Lead author and co-author of peer-reviewed publications, conference presentations, and technical reports.
- Contributor to standards development through ASTM and CGSB technical committees.

CURRENT POSITION

Fuels Chemist (CH-03)

September 2014 — Present

Characterization Laboratory, CanmetENERGY-Ottawa, Natural Resources Canada

Ottawa, Ontario

- Pioneer chromatographic, spectroscopic, and chemometric methodologies for characterization of petroleum products, renewable fuels, and biomass-derived materials, addressing analytical challenges for research groups and industry partners.
- Create novel chemometric methodologies adopted across multiple research projects.
- Develop open-source R tools to automate complex chromatographic and spectroscopic data analysis and visualization.
- Translate complex data into executive summaries and actionable technical reports for non-technical stakeholders and interdisciplinary teams.
- Lead and co-author journal publications and presentations, contributing to the dissemination of research findings.
- Contribute analytical design and technical expertise to research proposals, supporting departmental research initiatives.
- Serve as departmental scientific representative at national and international fora, including conferences, technical meetings, working groups, and standards boards.
- Champion data quality systems by serving as a CALA-certified ISO 17025 internal auditor, verifying analytical processes to ensure strict compliance to procedure and experimental reliability.
- Train and mentor junior staff and co-op students, fostering a culture of knowledge sharing and professional development.
- Act up as CH-04 with managerial and Sections 32 & 34 authority, as required.

Key Projects and Activities

Method Development Lead, SPME & Derivatization Analysis of Biorefinery Streams

2025 — Present

- Developed in situ derivatization and SPME workflows using Design of Experiments (DoE) principles to capture, stabilize, and quantify occupational health and safety priority carbonyl compounds in solid and liquid biorefinery streams.
- Established a cross-matrix analytical methodology enabling characterization of reactive carbonyl compounds across diverse biorefinery products and process streams.

Developer & Maintainer, PlotFTIR (<https://nrcan.github.io/PlotFTIR>)

2024 — Present

- Developed an R package for plotting and analyzing Fourier Transform Infrared (FTIR) and Raman spectroscopy data, enabling researchers to visualize and interpret complex spectral information effectively.

Author and Principal Investigator, International Round Robin Studies

2020 — 2025

- Led international round-robin studies of bio-liquefaction oils on behalf of Canada at IEA Bioenergy Task34.
- Produced two first-author papers, fostered international collaboration, and improved understanding of analytical methods.

Method Development Lead, GC-FID/MS Semiquantitative Analysis

2018 — 2022

- Built chromatographic method and custom data analysis algorithms to produce semiquantitative compositional reports with nearly 100% mass balance for complex samples.

- Validated method validity on petroleum, low carbon/renewable fungible fuels, and biofuel products.

PREVIOUS EXPERIENCE

Research Assistant (Co-Op)

2010 — 2011

DeNOx Group, CanmetENERGY-Ottawa, Natural Resources Canada

Ottawa, Ontario

- Investigated homogeneous catalysts for the reduction of NO_x in lean-burn diesel engine exhaust.
- Scaled catalyst synthesis by three orders of magnitude.
- Custom-built analytical instrumentation and analysis software to support research inquiries.

Laboratory Teaching Assistant

September 2011 — May 2014

University of Ottawa / University of Waterloo

Ottawa / Waterloo, Ontario

- Supervised undergraduate laboratories across analytical, organic, inorganic, and physical chemistry, instructing on advanced techniques and lab safety protocols.

EDUCATION

University of Ottawa

2012 — 2014

Master's of Science, Chemistry

Ottawa, Ontario

- Thesis: "Rhenium (I) Terdentate Compounds: Theoretical and Experimental Investigations"
- Seminar: "Recent Advancements in NO_x Abatement from Diesel Engine Emissions"

University of Waterloo

2007 — 2012

Bachelor's of Science, Honours Chemistry (Co-op), Music Minor

Waterloo, Ontario

- Thesis: "Solid Sample Analysis by Microplasma Optical Emission Spectroscopy"
- Achieved "Outstanding" co-op term ranking for each of the 5 work terms (2008-2012)

EXTRACURRICULARS & COMMUNITY LEADERSHIP

Clerk and Chair of Council (Executive Committee), Kanata Community Church

2024 — Present

- Progressed from Council Clerk to Chair, assuming increasing responsibility for governance, policy implementation, and organizational leadership.
- Lead council operations, supervise the Lead Pastor, and facilitate policy, financial, and organizational decisions on behalf of the congregation.

Author & Maintainer, fldataR R Package

2023 — Present

- Develop and maintain an open-source R package for Formula 1 data analytics, managing CRAN submission requirements, automated test coverage workflows, and community issue tracking on GitHub.

AWARDS & DISTINCTIONS

Scientific Innovation, Branch Award, CanmetENERGY-Ottawa

2026

- For the development of cross-matrix analytical methods for identification and quantification of occupational health and safety priority carbonyl compounds in biorefinery streams.

Excellence in Science, Departmental Award, Natural Resources Canada

2023

- For contributions to a Sustainable Aviation Fuel research and development project (The Sky's The Limit)

Positive Workplace Impact, Sector Award, Energy Efficiency & Technology

2022

- For participation in CanmetENERGY-Ottawa Emergency Response Team for medical and HAZMAT response.

Innovation & Creativity, Branch Award, CanmetENERGY-Ottawa

2021

- For the conception and development of laboratory database management tools, saving 0.25 FTE in the Characterization Lab.

Dean's Scholarship, University of Ottawa

2015

Dean's Honour Roll, University of Waterloo; University of Ottawa

2011-2014

Recognition of Collaboration, Departmental Award, Natural Resources Canada

2012

- For contributions to the development of a low-temperature selective catalytic reduction of NO_x with NH₃ in collaboration with Transport Canada.

PEER REVIEWED PUBLICATIONS

- Zhang, Y.; Monnier, J.; **Bulsink, P.**; *et al.* (2025). "Evaluation of diesel fuel production from bio-oils hydrodeoxygenation using unsupported MoS₂ catalysts" **Fuel Processing Technology**, 276, 108290. [10.1016/j.fuproc.2025.108290](https://doi.org/10.1016/j.fuproc.2025.108290).
- **Bulsink, P.**; *et al.* (2025). "Interlaboratory Study of Sample Homogeneity Impact on CHNS, Water, and ICP Analysis of Biomass Liquefaction Oils" **Energy & Fuels**, 39 (29), 14223-14236. [10.1021/acs.energyfuels.5c01309](https://doi.org/10.1021/acs.energyfuels.5c01309).
- **Bulsink, P.**; *et al.* (2020). "Results of the International Energy Agency Bioenergy Round Robin on the Analysis of Heteroatoms in Biomass Liquefaction Oils" **Energy & Fuels**, 34 (9), 11123-11133. [10.1021/acs.energyfuels.0c02090](https://doi.org/10.1021/acs.energyfuels.0c02090).
- **Bulsink, P.**; *et al.* (2016). "Capturing Re(I) in an neutral N,N,N pincer Scaffold and resulting enhanced absorption of visible light" **Dalton Transactions** 45, 8885-8896. [10.1039/c6dt00661b](https://doi.org/10.1039/c6dt00661b).
- Stanciulescu, M.; **Bulsink, P.**; *et al.* (2014). "NH₃-TPD-MS study of Ce effect on the surface of Mn-or Fe-exchanged zeolites for selective catalytic reduction of NO_x by ammonia" **Applied Surface Science** 300, 201-207. [10.1016/j.apsusc.2014.01.175](https://doi.org/10.1016/j.apsusc.2014.01.175).
- Stanciulescu, M.; Caravaggio, G.; Dobri, A.; Moir, J.; Burich, R.; Charland, J.-P.; **Bulsink, P.**; (2012). "Low-temperature selective catalytic reduction of NO_x with NH₃ over Mn-containing catalysts" **Applied Catalysis B: Environmental** 123, 229-240. [10.1016/j.apcatb.2012.04.012](https://doi.org/10.1016/j.apcatb.2012.04.012).

CONFERENCE & INVITED PRESENTATIONS

- **Bulsink, P.**; *et al.* (2026) "Characterization of Volatile Carbonyl Compounds in Biorefinery Streams using SPME-GC: Implications for Occupational Health and Safety" **Pyro2026 Conference**. Pisa, Italy
- **Bulsink, P.**; *et al.* (2025). "Multidimensional Chromatography of Bio/Renewable Energy Products " **Canadian Society of Chemistry Conference**. Ottawa, Ontario
- **Bulsink, P.**; *et al.* (2023). "Quantification of components without direct calibration by GC-MS/PolyArc®-FID" **American Chemical Society Conference**. San Francisco, California
- **Bulsink, P.** (2021). "Gas Chromatographic and Mass Spectral Analysis in Characterization Lab" *CanmetENERGY-Ottawa Science Seminar*
- **Bulsink, P.**; *et al.* (2020). "Results of the IEA Bioenergy Round Robin on the Analysis of Heteroatoms in Biomass Liquefaction Oils" *CanmetENERGY-Ottawa Science Seminar*

PROFESSIONAL AFFILIATIONS

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| • ASTM International | 2024 — Present |
| • American Chemical Society | 2019 — 2026 |
| • CGSB Petroleum Fuels Board | 2016 — 2025 |
| • Canadian Society for Chemistry | Periodic Member, 2013 — 2025 |